

EXPERIMENT – 3

SHELL SCRIPTING

1. Arithmetic Operations

```
``bash
#!/bin/bash

echo "Enter first number:"
read a

echo "Enter second number:"
read b

sum=$((a+b))
sub=$((a-b))
mul=$((a*b))
div=$((a/b))

echo "Addition = $sum"
echo "Subtraction = $sub"
echo "Multiplication = $mul"
echo "Division = $div"
``
---
```

2. Basic Shell Script Demo

```
``bash
```

```
#!/bin/bash
```

```
echo "Welcome to Shell Scripting"
```

```
name="Sanju"
```

```
echo "My name is $name"
```

```
echo "Enter your age:"
```

```
read age
```

```
echo "Your age is $age"
```

```
```
```

### 3. Fibonacci Series

```
```bash
```

```
#!/bin/bash
```

```
echo "Enter number of terms:"
```

```
read n
```

```
a=0
```

```
b=1
```

```
echo "Fibonacci Series:"
```

```
for (( i=0; i<n; i++ ))
```

```
do
```

```
    echo -n "$a "
```

```
fn=$((a+b))  
a=$b  
b=$fn  
done  
````  

```

#### 4. Leap Year Program

```
````bash  
#!/bin/bash  
  
echo "Enter year:"  
read year  
  
if [ $((year % 400)) -eq 0 ] || [ $((year % 4)) -eq 0 ] && [ $((year % 100)) -ne 0 ]  
then  
    echo "$year is a Leap Year"  
else  
    echo "$year is not a Leap Year"  
fi  
````  

```

#### 5. Reverse a Number

```
````bash  
#!/bin/bash
```

```
echo "Enter a number:"
read num

rev=0

while [ $num -gt 0 ]
do
    rem=$((num % 10))
    rev=$((rev * 10 + rem))
    num=$((num / 10))
done

echo "Reversed Number = $rev"
...
```

Steps to Run Shell Script

Create File

```
```bash
nano file.sh
...

```

##### ## Save File

Press:

```
```bash
```

CTRL + X

Y

ENTER

```
...
```

#Give Permission

```
```bash
```

```
chmod +x file.sh
```

```
```
```

#Run Program

```
```bash
```

```
./file.sh
```

```
```
```